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Cayenne

Design empathy in the digital age

An independent, non-commercial white paper publication examining serviceability, the design empathy framework, long-term user outcomes, and the software-defined vehicle.

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“You can’t understand good design if you do not understand people; design is made for people.”

–Dieter Rams

Table of contents

Superscript key

#	Citation number
G	Glossary entry

Introduction

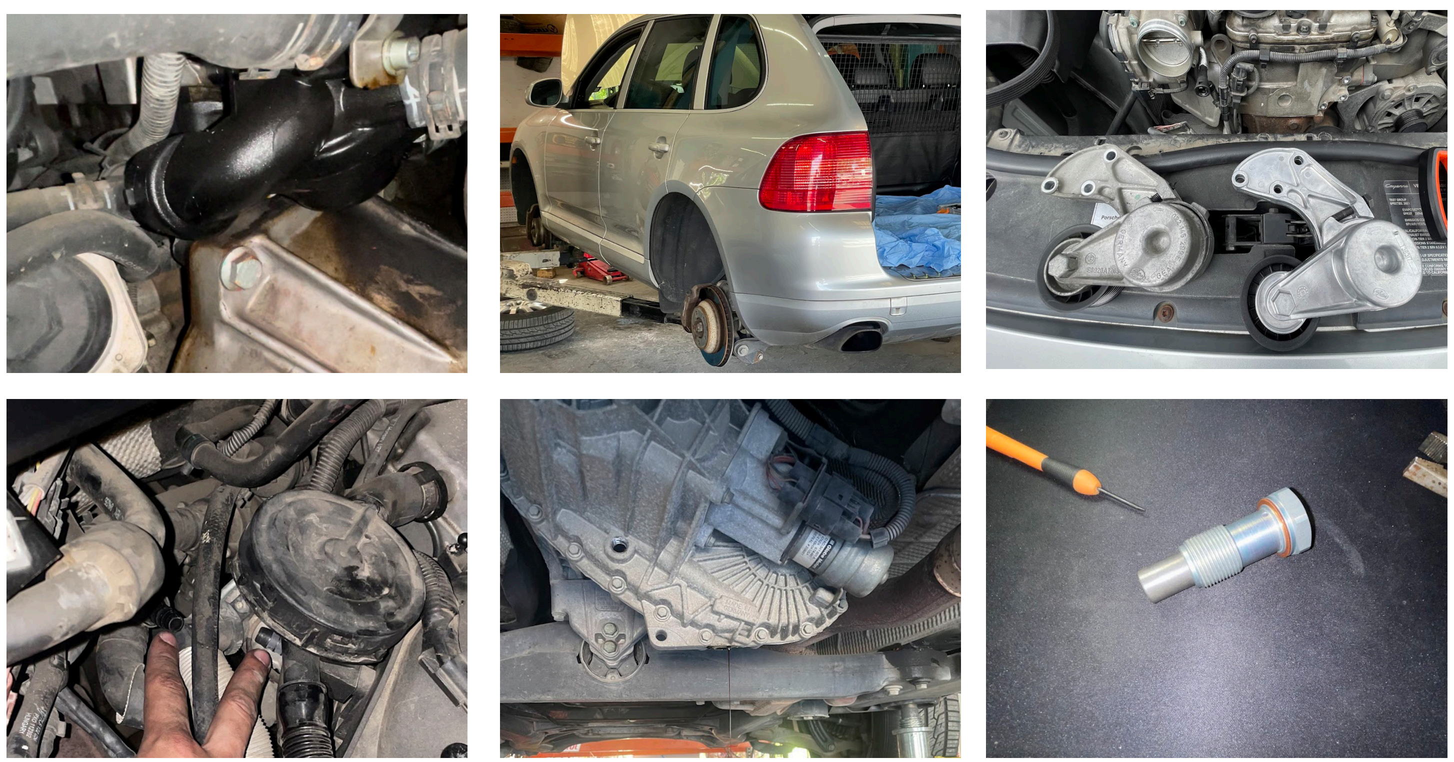
The subject vehicle was purchased by the author in July 2024 as an automotive project with no prior mechanical experience and the goal of turning it into a daily driver. The experience brought about a desire to understand how design decisions affect long-term ownership viability. The Porsche Cayenne was selected for its versatility, excellent parts availability, decent serviceability for the product category, and a robust enthusiast knowledge base. While service documentation was either difficult or expensive to obtain, this enthusiast community has created guides that compensate for information gaps with tribal knowledge.

Over two years and 11,000 miles (17,700 km), the author performed numerous repair and maintenance procedures, documenting component failures, costs, and serviceability challenges. This analysis draws directly from that hands-on experience, supplemented by manufacturer documentation, regulatory archives, and industry research.

Purchase price: $3,000 (USD)
Mileage at purchase: 175,050 miles/281,716 km
Distance driven: 11,000 mi/17,700 km
Fuel cost: $0.24/mi, $0.15/km
Total operating cost: $1.16/mi, $0.72/km
DIY service time invested: Approx. 40 hours

▶ Braking news

Zane in a formerly white T-shirt after completing a four-corner brake service with his friend Lynette, including torquing caliper mounting bolts to manufacturer spec of 200 ft-lbs/bolt (271.2 Nm).



▲ Service smörgåsbord

From top left to bottom right: Thermostat housing above transmission, brakes, serpentine belt and tensioner, brake booster vacuum line, transfer case, timing chain tensioner (post-assembly).

▲ Are you lost?

In the background, an exclusive German SUV looms over a random Ford.

◀ Peer reviewed

Project oversight and rigorous peer review process conducted by Kitty.

Abstract

What happens when a moonshot product ages into a used car?
This white paper focusing on serviceability and the software-defined vehicle (SDV) seeks to answer that question using a 21-year-old first generation Porsche Cayenne and the design empathy framework. A software-defined vehicle is an automobile where major functions are controlled, managed, and enhanced by software, with wireless and over-the-air (OTA) update capabilities. As cars become more complex and resource intensive to create, purchase, and maintain, serviceability becomes as critical as component lifespan. The first generation Porsche Cayenne represents a transitional period where service was still relatively accessible, but manufacturers externalized costs to users through design choices that prioritized cost and production efficiency over long-term serviceability. This paper explores these failures and the pattern of industrial design neglect they show historically and concurrently, then proposes solutions to build better vehicles for the future.

The design empathy framework

Design empathy is the practice of anticipating a product's complete impact on users across its entire lifecycle; from acquisition through daily use, failure, repair, and end-of-life. Empathetic design assumes products will be used in unforeseen ways, outlive their warranty periods, change hands, and eventually fail. It minimizes harm, cost, and complexity when those inevitable events occur, rather than externalizing these consequences to users. This keeps products in use longer, reducing environmental impact and maintaining the brand's reputation across ownership cycles.

In an ideal model, good design empathy ensures that products do not exploit and burden the end user. This framework as a developmental tenet makes a product not only desirable in its prime, but viable long after. Brand, environment, and user all benefit from a long tail of ownership that anticipates failure and provides accessible means to remedy it, whether physical or digital. We are forced to reckon with what endurance and value mean for end users today when vehicles outlive their design briefs (and in some cases, their makers) by decades.

As the automotive industry enters the digital age, this pattern of neglect has evolved in scope and scale. Components that used to be difficult to service a decade ago are becoming impossible to address independently as they are now software locked, built into larger assemblies that have to be replaced in full, and externally dependent on their manufacturers.

Given the current status quo established in latter parts of this paper, one can infer that the mechanical vehicle of yesterday and today's SDVs sit on different ends of the user agency spectrum. Modern cars increasingly rely on software for basic functions such as braking, steering, and throttle, but what happens when these cutting edge systems are compromised? Physical failures are now just one consideration, as the introduction of the SDV has introduced failure modes that affect entire fleets simultaneously.

This shift mirrors a broader trend in consumer goods. Right-to-repair has become necessary precisely because manufacturers have adopted planned obsolescence[G] as standard practice, even for products that represent significant financial investment and environmental impact. The automobile, once the standard for durable consumer goods (designed for decades of use, multi-generational ownership, and aftermarket support), is following this trajectory.

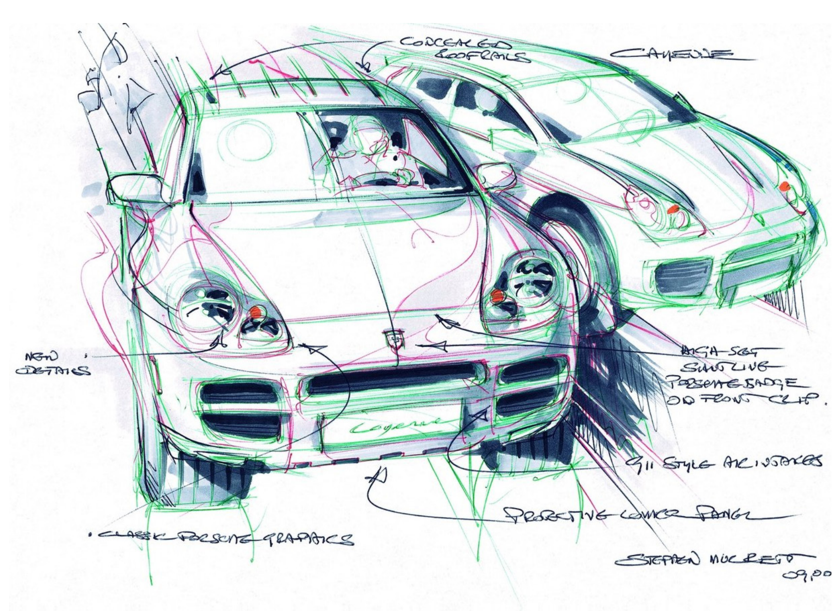
Product context

The Cayenne was conceived to save Porsche from the brink of bankruptcy. Porsche was enduring a financial crisis in the 1990s due to an aging lineup, a strong Deutschmark that made exports expensive, and increasing competition from other brands. The introduction of their compact Boxster sports car in the mid-1990s helped alleviate their financial situation, but sports cars remained a niche category. Larger profits were attainable by targeting volume sales in a new segment that was taking the market by storm: The sport utility vehicle (SUV).

Under codename Project Colorado, Porsche and Volkswagen co-developed a new SUV platform to be utilized by both brands (and eventually, a third: Audi). Once platform basics were established, the brands parted ways for continued development of their respective models. Porsche developed new powertrains and re-engineered suspension components to help the vehicle resonate more closely with Porsche buyers, but also used a slightly modified Volkswagen 3.2 VR6 engine in the base model to cut costs and expand volume appeal.

▶ Familiar form

Original sketches by Design Manager Stephen Murkett. The Cayenne was designed to channel the styling of their well-known sports cars to an SUV form factor. The execution was notably polarizing for the era.



Original image © Dr. Ing. h.c. F. Porsche AG.

▼ A divisive addition

In a lineup of sports cars, Porsche's first four-door production vehicle was an outlier that angered brand purists. It would later become the brand's bestselling model.

Original image © Dr. Ing. h.c. F. Porsche AG.

The risk of this product's entry to market was in the perception and context of their brand at the time. Porsche was exclusively known as makers of sports cars, so the Cayenne SUV was seen by enthusiasts as an affront to the brand's heritage. Out of fear of not being taken seriously, Porsche developed the Cayenne lineup with a wide spectrum of use cases in mind (on road, off road, and on the track). While initially controversial, markets responded warmly to the vehicle after launch.

This sentiment was reflected in sales figures; the Cayenne became Porsche's bestselling model by 2007, eclipsing the historic 911 sports car that was the brand's bread-and-butter, and proving Porsche's venture into SUVs was well-founded and highly profitable. The vehicle is credited with redefining the SUV segment, proving high-end luxury car makers can build profitable SUVs, a blueprint now followed by the likes of Aston Martin, Ferrari, Lamborghini, and Rolls-Royce.

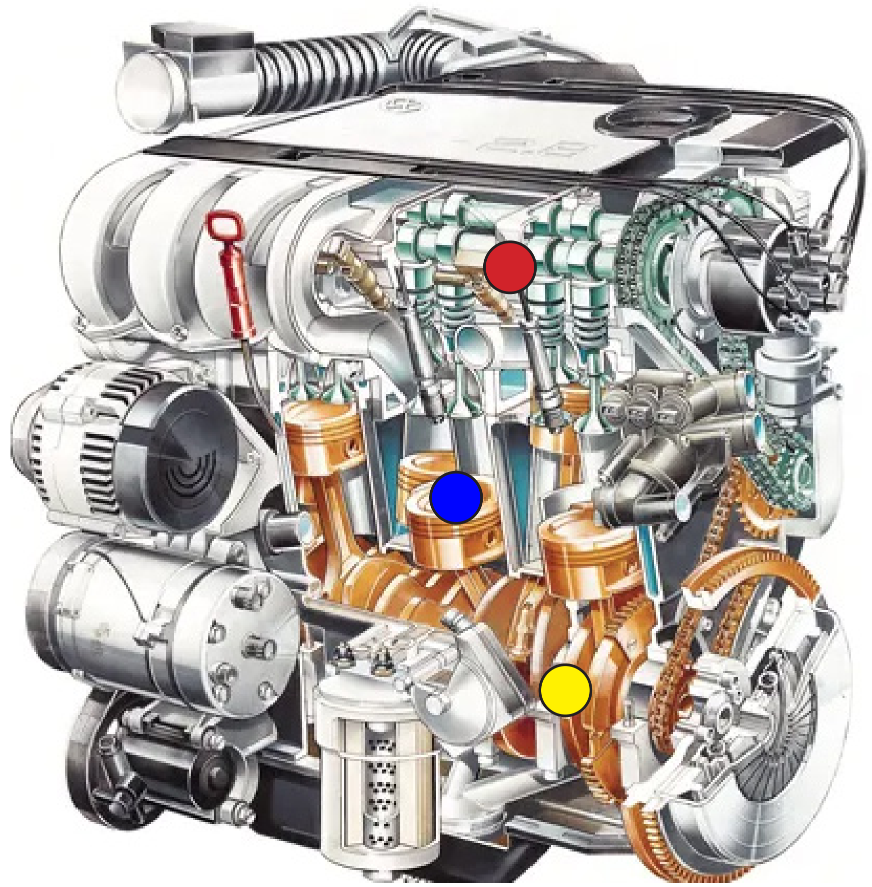
Volkswagen VR engines

▶ **Technically brilliant**

In the world of engines, the VR6 was an unusual creation that merged the simplicity of an inline layout with the compactness of a V layout. It was heavy, smooth, and had a characteristic growl.

Diagram key

- Top end
- Cylinders
- Bottom end



2.8 VR6 engineering cutaway (1991) - Original image © Volkswagen AG

The VR6 was introduced by Volkswagen in 1991 to solve a packaging problem. Standard six cylinder engines are too long (Inline) or too wide (V-shaped) to fit in small cars, so Volkswagen engineered the VR6 to utilize a novel staggered design. By zig-zagging the cylinders at a narrow 15° angle, they created a ‘Short V’ slim enough to fit in compact engine compartments originally designed for four cylinder engines, and it only required one cylinder head, unlike traditional V-engines that required two due to the larger angle. The result was a simple, power-dense, and robust powertrain with a wide array of applications, from industrial equipment to performance vehicles. In the late 1990s, Volkswagen engineers coupled two VR engines at the bottom-end, developing the W-shape that became the foundation for the power-dense, high-cylinder-count powertrains found in the lineups of Audi, Bentley, Bugatti, and Volkswagen’s own throughout the 2000s.

Linde H50

Volkswagen Phaeton

Volkswagen Sharan

Audi TT

Bugatti Veyron

Volkswagen Passat

Mercedes-Benz V280

Volkswagen Golf

Porsche Cayenne

Artega GT

Volkswagen Beetle

Volkswagen Atlas

Audi Q7

Škoda Superb

Volkswagen Corrado

Bentley Continental

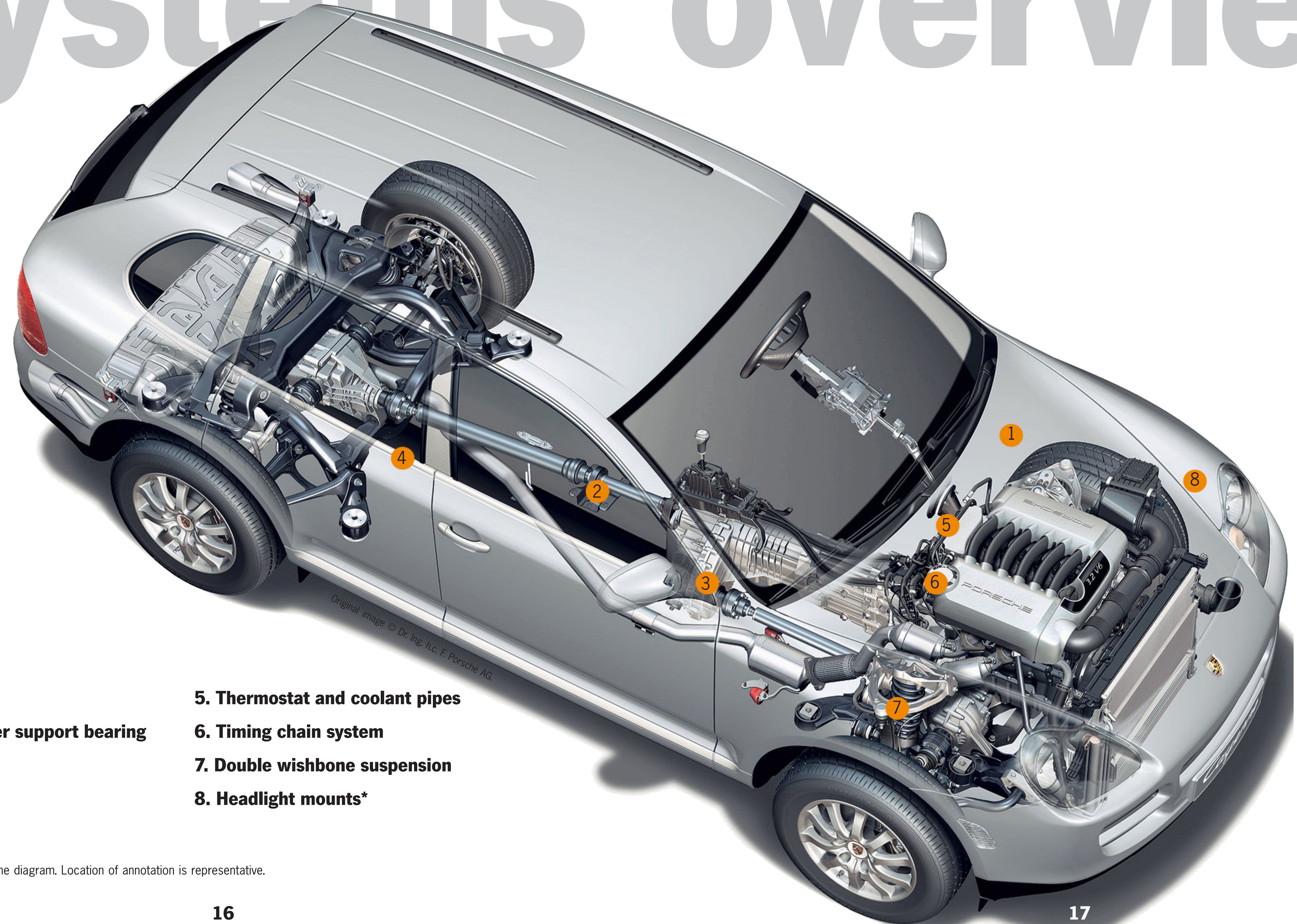
Bentley Bentayga

Volkswagen W12 Concept

▲ **Family reunion**

Volkswagen’s VR and W engines have been used in various applications over the years.

Systems overview



1. **Rubber drains***
2. **Driveshaft center support bearing**
3. **Transfer case**
4. **Fuel pumps***
5. **Thermostat and coolant pipes**
6. **Timing chain system**
7. **Double wishbone suspension**
8. **Headlight mounts***

*Component is not visible in the diagram. Location of annotation is representative.

System vulnerabilities

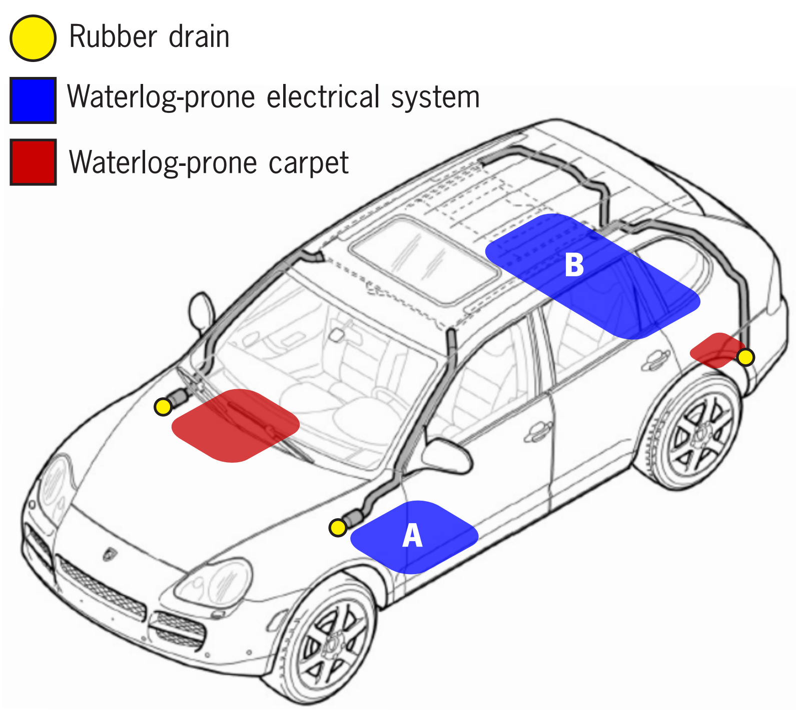
Rubber drains

This system acts as a rain gutter, giving water a path through and out of the vehicle. The first generation Cayenne was designed with off-road use in mind, necessitating the use of a seal system to keep water out during events such as river fording. This seal system involves a number of rubber drains with ends that taper into a plus shape. These ends were designed to guide water out without letting organic matter back in.

Over time, the rubber drains harden and clog with organic material, allowing water to pool in the cabin and around important electrical systems. The service manual does not advise replacing these drain plugs. Neglecting this water buildup can cause a slew of electrical issues that can immobilize the vehicle. This issue plagues the entire Cayenne lineup, as well as other vehicles using this platform.

▶ When it rains, it pours

This graphic illustrates a portion of the drain system, as well as some waterlog prone areas in the car's cabin. Red zones are carpeted areas not within the immediate vicinity of important electrical systems, in contrast to blue zones: A houses the vehicle's main wiring harness (handling the CANBUS[G]) beneath the carpet, whereas B houses audio and air suspension components.

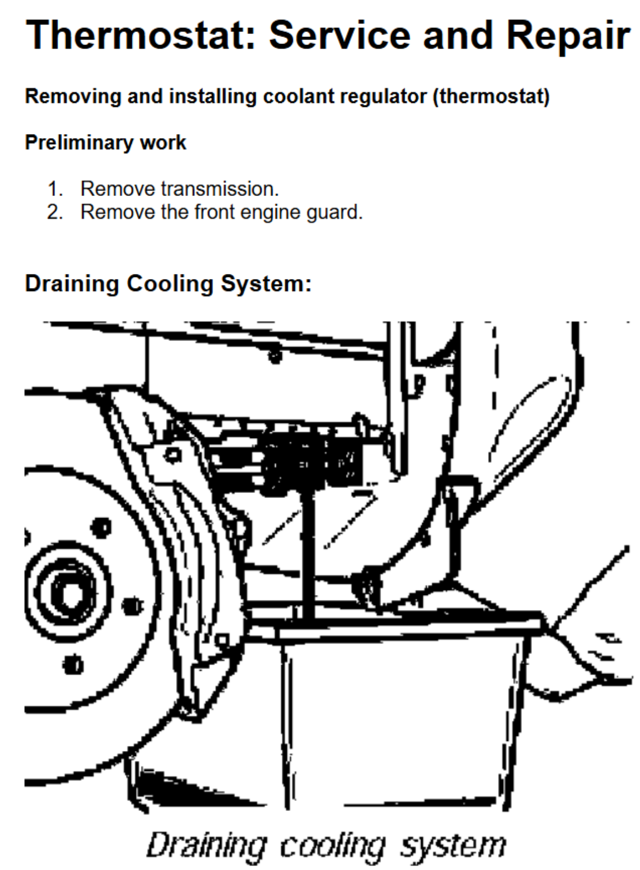


Original image © Porsche AG, annotations are original

Coolant pipes

These are typically made of glass filled nylon, an engineering-grade plastic, used in a sealed engine compartment (which is highly thermally dense due to the VR6's compact design), and located in hard-to-access areas. Deterioration of these plastic parts during normal use is expected due to heat cycling, but accessibility and specific material choices do not reflect this reality. Volkswagen has had this issue surface on multiple VR6-equipped vehicles since its introduction.

This has earned a particularly vulnerable coolant distribution pipe the enthusiast nickname 'crack pipe' for its propensity to randomly burst as vehicles age. Due to the system's location on the subject vehicle, a thermostat or coolant pipe replacement (typically a straightforward job for an approximately $100 part) requires removal of major components as the first step. Porsche's choice not to utilize aluminum for specific portions of their VR6 variant (such as the coolant distribution pipe) shows a misalignment between cost efficiency and long-term operability.



Thermostat: Service and Repair

Removing and installing coolant regulator (thermostat)

Preliminary work

1. Remove transmission.
2. Remove the front engine guard.

Draining Cooling System:

Draining cooling system

Original image © Porsche AG

◀ Step 1: Remove... transmission?

This is a diagram from Porsche's official service documentation for the Cayenne VR6. The location of key cooling system components between the firewall and the back of the engine, a result of repurposing a transverse-optimized engine for longitudinal use, makes otherwise straightforward repairs require major disassembly. This accessibility is the VR6 model's main achilles heel.

▶ **The cost of poor empathy**

The subject vehicle needed a thermostat replacement, a job that exceeded the author's mechanical abilities. In order to replace these simple components, the transmission had to be removed from the vehicle. The shop quote shows the large delta between parts and labor costs for what should be a simple job.

Labor		Total
REPLACE THERMOSTAT		$2,640.00
Part	**Qty**	**Total**
Metrix Engine Coolant Thermostat / Water Outlet Assembly	1	$173.91
PINK COOLANT	3	$105.00
Engine Coolant Pipe	1	$75.90
		Job Total $2,994.81

Timing chain system

The job of the timing system is to synchronize rotation of the crankshaft and camshaft(s), ensuring engine valves open and close at the correct time in relation to the piston movement; failure of this system is guaranteed failure of the entire engine. Like the thermostat, the timing system on the VR6 engine is located on the back, a result of its original development for transverse use.

Service of this system on the VR6 model requires removing the entire engine from the car, as opposed to the clean-sheet design of Porsche's eight cylinder engines used in higher end models, which had timing components on the front. While the chain is considered a "lifetime" part by Volkswagen, chain stretch and guide wear are inevitable results of use and will have to be serviced to avoid total engine failure.

Headlight mount bracket

Located in the engine bay, the plastic frame that holds the headlights in place warps over time and causes an inability to hold the headlight in place. Aside from heat and age related issues, the latch design is theft-prone. It is accessible from underneath the headlights, even with the hood closed.

This has made these cars massive targets for headlight theft with little to no effort or skill needed. Porsche designed this mechanism to allow for easy replacement of the headlight bulbs by making the headlights slide out when unlatched, but they inadvertently also made them extremely easy to steal.

> **"Porsche owners beware – cannabis growers may steal your headlights**
> A spate of thefts in Amsterdam suggests the Porsche's xenon lights could be the perfect bit of horticultural kit for cultivating marijuana"
>
> **–The Guardian,** December 2012

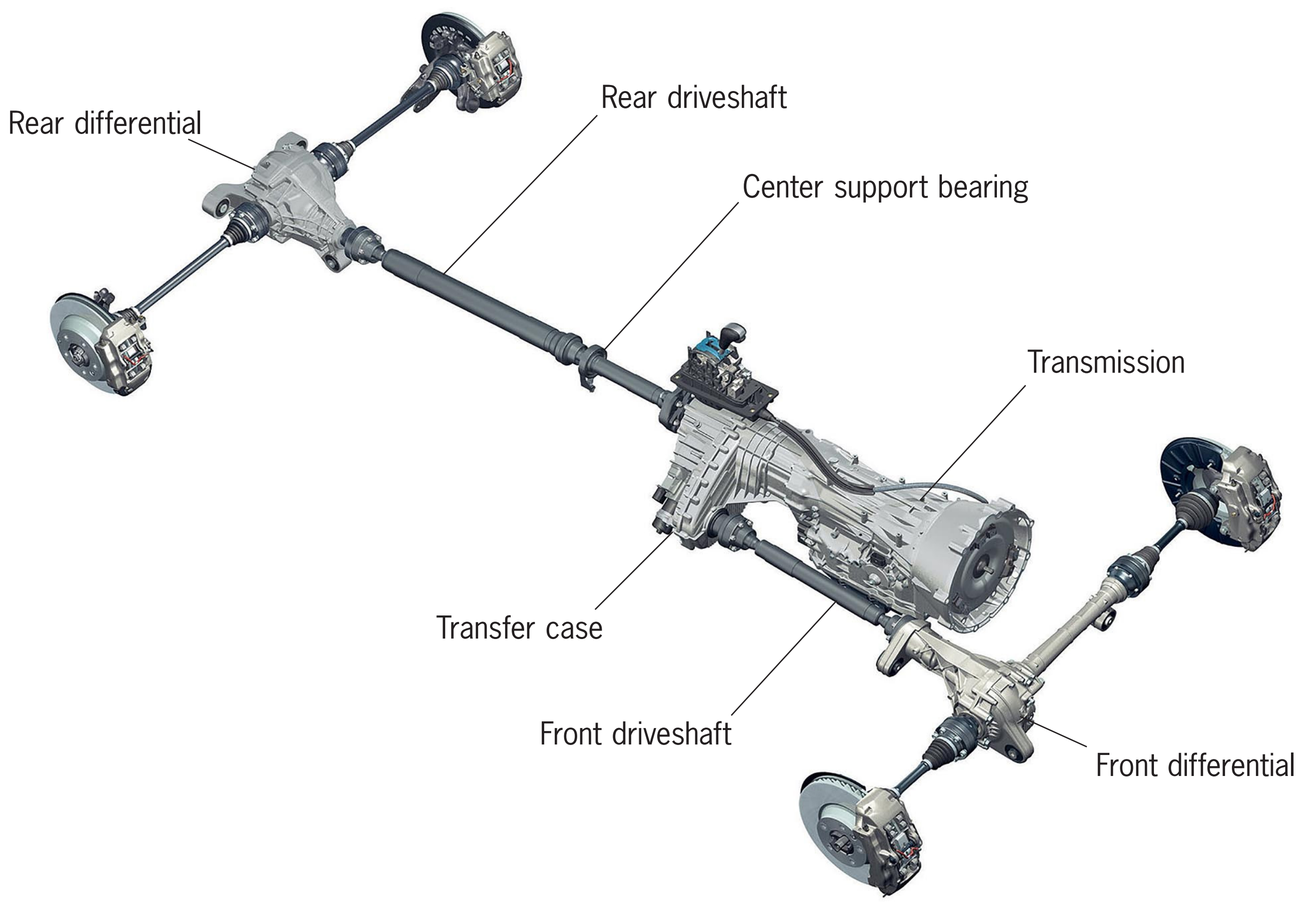
◀ **Shoelace engineering**

Due to heat-induced warpage of the mount brackets, the author had to wield the car's design flaw to remove the headlights for a routine bulb replacement. Using the ample gap between the headlights and bumper, he undid, then later re-attached, the latch securing the headlights to the mount by maneuvering a shoelace wrapped around it.

System strengths

While some components exhibit accessibility problems or poor material choice, others are designed thoughtfully and contribute to the vehicle's potential for longevity. Some components still have notable weak points, but their serviceability allows for proactive intervention.

Key drivetrain components



Original image © Porsche AG, annotations are original

Fuel pumps: When these components fail, access is simply a matter of lifting up the hinged rear seat assemblies and cutting an aperture in the carpet under both outboard rear seats, gaining access to two service ports that lead to the fuel pumps. This is in contrast to some other vehicles that require the removal of the entire fuel tank to simply access the pump(s).

Double wishbone suspension: Coupled with the vehicle's weight of approximately 4,762 lbs (2,160 kg), and the resulting suspension geometry from the car's sporting/all terrain ambitions, this car's suspension system takes a beating. Control arm bushings wear out around 100k miles/160k km of normal use. Neglect results in sloppy handling, meandering, and poor ride quality. This is an excessive wear item as opposed to a design flaw, but access and remedy is relatively accessible. Higher trim levels are equipped with air suspension, an option that introduces an expensive range of new failure points. The subject vehicle is not equipped as such, utilizing a standard coil-sprung setup as shown in the systems overview.

Driveshaft center support bearing: The Cayenne utilizes a two piece rear driveshaft setup that is common in larger vehicles where a longer driveshaft would cause unwanted NVH[G] and packaging issues. This necessitates the use of a carrier/support bearing, which is designed to dampen driveshaft movement and vibration during operation. Over time, excessive wear on the driveshaft support bearing due to rubber degradation results in significant noise and vibration during operation, but this component is easily accessible from beneath the car.

Transfer case: This component is the heart of the vehicle's all-wheel-drive system and contributes directly to the vehicle's diverse performance profile by variably distributing power between both axles. The transfer case is extremely easy to service; the fill and drain plugs are both easily accessible on the unit from the undercarriage. Porsche's long service intervals suggested for this component have caused transfer case failures en-masse on multiple models that resulted in class action litigation against them.

The 2000s bore witness to a new practice in the automotive industry.

Citing technological advancements in manufacturing and fluid composition, carmakers increased service intervals for components on their vehicles. This is a marketing strategy used to artificially lower a vehicle's running costs as advertised, making models look more cost-competitive and low-maintenance on paper. In the industry, "lifetime" often means the length of the vehicle's warranty period[3]. These extended service intervals result in more frequent major component failures due to fluid degradation that causes sludge buildup, oil consumption, and friction between powertrain components[2].

PORSCHE

Minor Maintenance Checklist – Cayenne/S/T (2003-06)
Required Maintenance and Lubrication Service

Additional Maintenance every 160,000 miles / 240,000 km or 16 years

- ❏ Change front differential oil
- ❏ Change rear differential drive oil
- ❏ Change transfer case oil
- ❏ Change manual transmission oil
- ❏ Change Tiptronic transmission oil and ATF filter

Porsche PDF reference # PNA 000 162 CA - Cropped for brevity

▲ **The neglect protocol**

Porsche's maintenance checklist suggests transfer case service every 160k mi/240k km. Specialists suggest 20-60k mi/32-96k km intervals[8,9], 3-8x more frequently than Porsche.

In the gap between what a product promises and what it delivers lies the truth about what a company values.

The automobile, much like any product, encodes the priorities of its maker. The vehicle referenced in this paper provides insight into the corporate structures and thinking that necessitate the conception of volume products.

The historical factors that shaped Porsche's Cayenne are now irrelevant, but the corporate interests that influenced its design decisions at the turn of the millennium have not changed. They are now simply shaping different products with new problems at a larger, more complex scale.

The following section explores the evolution of this pattern and how these modern failures are qualitatively different to those of the past.

Evolution

Historically, automakers have found that the consequences of poor design are forgotten long after the appeal of cost-effective business decisions are. These examples showcase how the practice of externalizing risk to owners has evolved over the decades. Among others, the Ford Pinto's case still holds relevance due to its conception in a regulatory vacuum, a problem we currently face as SDV development accelerates faster than legislative institutions can keep up.

1965 | Chevrolet Corvair

◀ 'Unsafe at Any Speed' published

Ralph Nader's infamous report exposed how manufacturers prioritized profit over safety and got away with it, focusing on the Chevrolet Corvair, GM's attempt at a rear-engined Beetle competitor with a deadly propensity to spin out.

1981 | Ford Pinto

◀ Grimshaw v. Ford ruling

A California court confirms Ford knew the Pinto's rear-mounted fuel tank would kill people in the event of a rear-end collision and chose not to fix it because it was not profitable to do so.

2009 | Toyota Camry

◀ Runaway Toyota vehicles

After reports of vehicles accelerating uncontrollably and a deadly crash, Toyota recalled about 8 million vehicles for floor mats that would trap the accelerator pedal. They later admitted to concealing a separate issue regarding 'sticky' pedals, resulting in a $1.2 billion criminal penalty[15, 16].

2014 | Chevrolet Cobalt

◀ GM faulty ignition switches

GM knowingly used faulty ignition switches for over a decade in their compact cars while concealing the issue. These switches disabled vehicles while in motion, deactivating steering, braking, and airbags. 124 people were killed as a result, and GM paid $900 million in settlements[17, 18].

2024 | Fisker Ocean

◀ Fisker collapses

Fisker Inc. files for bankruptcy. 11,000 mechanically sound vehicles are orphaned when Fisker's cloud servers go down. Their core functions were locked behind cloud-based software only Fisker controlled.

2025 | Porsche 911

◀ Mass-immobilized Porsches

Hundreds of Porsche vehicles across Russia stop working. Factory-installed tracking module lost satellite signal, interpreted the outage as theft, and immobilized the vehicles.

Design Ethics 101

Weak links in a product's design will always surface; manufacturers have a duty to ensure that these links do not compromise a product that users trust. In 2009, Toyota faced massive backlash after knowingly selling cars with 'sticky' accelerator pedals. After a deadly crash, Toyota initially recalled 8 million vehicles for floor mats that could trap the accelerator, while concealing defective 'sticky' pedals that would jam partially depressed due to material wear.

Toyota paid a $1.6 billion class action settlement in 2013, and a separate $1.2 billion criminal penalty to the DOJ in 2014 after admitting to misleading the public and regulators[15, 16].

Source: San Diego County Sheriff's Department / NHTSA

▲ **Criminal negligence**

The wreckage of a 2009 Lexus ES350 in San Diego, California after a deadly crash caused by a floor mat trapping the accelerator pedal. This accident sparked a legal and corporate crisis for Toyota that culminated in a $1.2 billion criminal settlement.

Source: 1978 NHTSA Crash Test Film (held by National Archives, RG 416)

◄ **Unsafe at any speed**

Images from the 1978 NHTSA crash test of a 1971 Chevrolet Impala rear-ending a 1972 Ford Pinto[7]. The Pinto's rear-mounted fuel tank ruptures then bursts into flames upon impact.

The 1970 Ford Pinto's rear-impact fuel-tank vulnerability is another example of this negligence, where layout choices had deadly consequences that Ford was aware of. At the time, Ford was tasked with cheaply producing a competitive compact vehicle under an aggressive development timeline. The resulting Pinto utilized a fuel tank that sat aft of the rear axle, between that and the rear bumper, which posed a very high risk of spontaneous combustion in the event of a rear-end collision. Ford knew about this liability, but their internal cost-benefit analysis concluded that the cost of a redesign was more substantial than the cost of legal settlements for families who had lost loved ones[6].

Today, the pattern of externalizing risk to users persists and accelerates, albeit in different forms. Manufacturers no longer make cost-benefit calculations about physical safety (as regulatory oversight largely prevents this), but long-term ownership viability now hangs in the balance. Software-defined vehicles introduce dependencies that render products artificially obsolete when manufacturer support ends, externalizing the existential risk of discontinuation or failure entirely to customers. While the cost-benefit analysis has shifted from 'recall costs vs. settlement costs' to 'engineering robust systems vs. capturing recurring revenue', the outcome remains the same. Just like Ford and Toyota's failures were egregious enough to lead to regulatory reform, SDVs are only beginning to face similar scrutiny.

Software-defined vehicles

Automobiles are inherently political products; vehicles of any given era are a reflection of the social, political, and technological climate of the time. 1950s post-war optimism gave rise to extravagant American vehicles, whereas the 1980s computer-aided design boon allowed manufacturers to engage with previously impossible aerodynamic shapes. The inverse is also true; political and technological tumult can also impact products.

As carmakers' products embrace the digital age, buyers will experience growing pains. This was true in the 1970s when abrupt demand for low cost and fuel efficient transport resulted in poorer quality vehicles, as well as the early 1910s, just as mass production was starting to proliferate and cars were still extremely unreliable. Both periods were massive inflection points for the automotive industry that have shaped its trajectory since. We are facing another inflection point in the 2020s, but this time surrounding environmental obligations and the software-defined vehicle.

The physical issues with the Porsche Cayenne presented in this paper, a relic compared to modern vehicles, are now just one facet when juxtaposed with the array of continuously evolving failure points in the digital realm. The Cayenne's inaccessible thermostat is frustrating and expensive, but it can be fixed. A shop or determined owner with the right tools can remove the transmission and replace the part; the knowledge exists in forums, parts are available from suppliers, and the work is possible, even if it is difficult.

Cars built today are increasingly reliant on connectivity and telemetry at a scale that didn't exist just 15 years ago. Manufacturers now market features such as over-the-air (OTA) updates, semi-autonomous driving, and remote functions through smartphone applications that rely on internet connectivity to function, introducing new liabilities to the user experience in the process. SDVs introduce failure modes that differ fundamentally from traditional automotive problems. This paper groups these failure modes into three categories:

Accidental: Software bugs that disable cars after OTA updates.

Structural: Data breaches, external dependencies. A result of inadequate regulation.

Intentional: Planned obsolescence, surveillance, invasive telemetry[G] at users' expense.

These categories and their examples show how different kinds of design empathy failures manifest, as well as their impact on both user and manufacturer. Some examples also highlight how a design empathy failure can actually be considered a success to the manufacturer when measured by different metrics: revenue and profit. Certain failure modes can be categorized in more than one way, such as external dependencies that can be both structural and intentional.

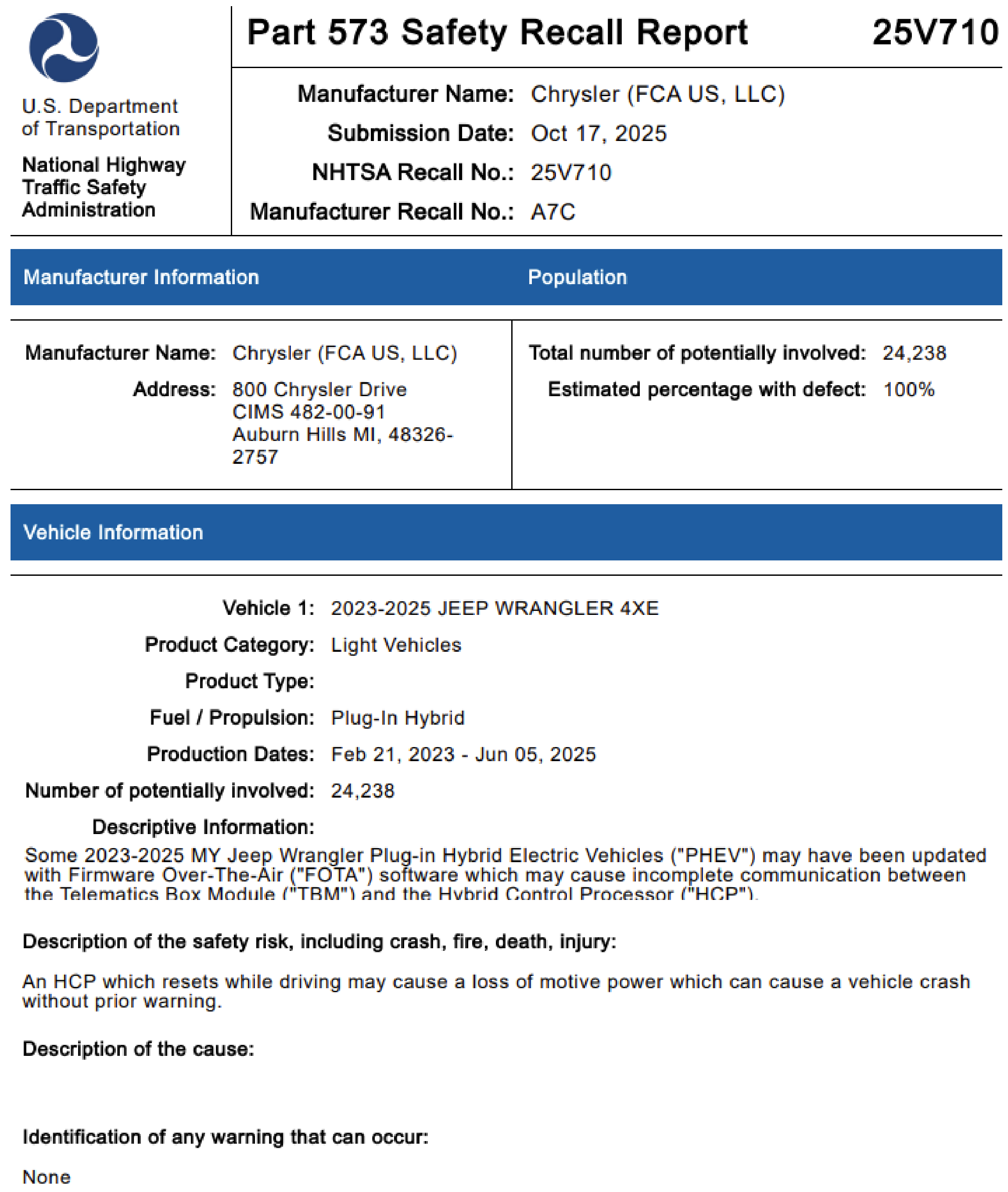
New feature: Brick mode

In 2025, Jeep rolled out an OTA update to hybrid Wrangler 4xe vehicles in their lineup. Due to severe software bugs included in the update, the vehicle's telematics box module (TBM) lost communication with the hybrid control processor (HCP). This misconfiguration causes the HCP to restart and eliminate power in the process, causing vehicles to become immobile and strand their owners, with some people even experiencing their vehicles lose power and shut down at highway speeds.

Jeep representatives went to forums to instruct customers to defer the update if they can, assuring them that Jeep is working on a fix. Aside from the unintended powertrain failures, the update was not designed to alter other vehicle functions, prompting Jeep to revert to a prior software version as a temporary emergency 'update' while they worked on providing a solution to distraught customers.

According to NHTSA's recall report, approximately 24,000 Wrangler 4xe vehicles were affected, with Jeep estimating that the entire production run was impacted. With a failure rate this comprehensive, one can reasonably assess that minimal quality control work has been conducted on software with the potential to impact powertrain systems.

> **"Please exercise extreme caution this evening if you have completed the update.** If you have NOT completed the update and see the pop-up, please continue deferring so that the update does not go through."
>
> **–Kori,** JeepCares support representative[10], 4xe Forums (October 2025)



U.S. Department of Transportation

National Highway Traffic Safety Administration

Part 573 Safety Recall Report **25V710**

Manufacturer Name: Chrysler (FCA US, LLC)
Submission Date: Oct 17, 2025
NHTSA Recall No.: 25V710
Manufacturer Recall No.: A7C

Manufacturer Information

Manufacturer Name: Chrysler (FCA US, LLC)
Address: 800 Chrysler Drive CIMS 482-00-91 Auburn Hills MI, 48326-2757

Population

Total number of potentially involved: 24,238
Estimated percentage with defect: 100%

Vehicle Information

Vehicle 1: 2023-2025 JEEP WRANGLER 4XE
Product Category: Light Vehicles
Product Type:
Fuel / Propulsion: Plug-In Hybrid
Production Dates: Feb 21, 2023 - Jun 05, 2025
Number of potentially involved: 24,238
Descriptive Information:
Some 2023-2025 MY Jeep Wrangler Plug-in Hybrid Electric Vehicles ("PHEV") may have been updated with Firmware Over-The-Air ("FOTA") software which may cause incomplete communication between the Telematics Box Module ("TBM") and the Hybrid Control Processor ("HCP").

Description of the safety risk, including crash, fire, death, injury:

An HCP which resets while driving may cause a loss of motive power which can cause a vehicle crash without prior warning.

Description of the cause:

Identification of any warning that can occur:

None

▲ Loss of motive power

The NHTSA's recall report for the Wrangler 4xe highlights the unanticipated, dangerous nature of the OTA update's ability to disable a moving vehicle.

Surprise backdoors

Manufacturers today have the power to control your vehicle remotely with no input from the driver required. Tesla showed that this is possible in January 2025 when a Cybertruck was used in a bombing on the Trump International Hotel in Las Vegas. In a televised press conference, Las Vegas Sheriff Kevin McMahill publicly thanked Elon Musk for his collaboration in the investigation, stating that the Cybertruck was locked due to the force of the blast, and that Tesla was able to remotely unlock the vehicle to aid law enforcement efforts.

This incident reveals both sides of centralized vehicle control. In this case, the capability aided a legitimate law enforcement investigation, but the same root access that enables Tesla to unlock vehicles or provide data for police also means manufacturers (and bad actors who compromise their systems) can access vehicles without owner consent. A backdoor is a backdoor, regardless of who uses it.

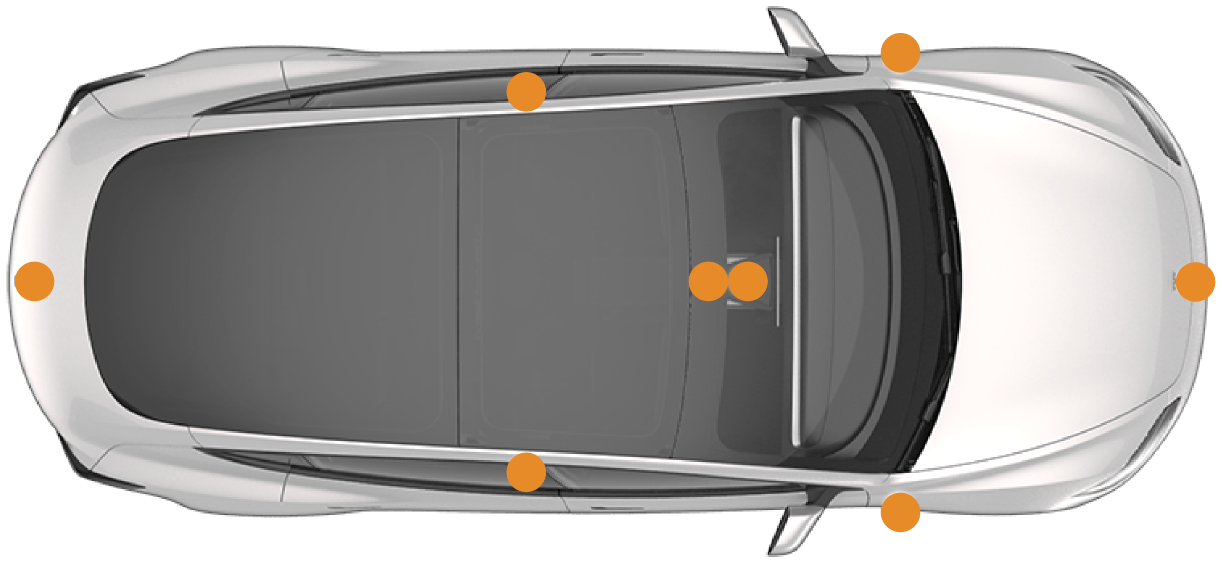
This vulnerability is not exclusive to Tesla. As far back as 2015, security researchers Charlie Miller and Chris Valasek remotely exploited a 2014 Jeep Cherokee's cellular-connected infotainment system from a laptop[14], gaining the ability to kill the engine at highway speed, disable brakes, and control steering, door locks, and the transmission. This resulted in Fiat Chrysler, Jeep's parent company, issuing a recall to update infotainment software for about 1.4 million vehicles.

▶ **Trail Rated**

The Jeep Cherokee with Wired journalist Andy Greenberg behind the wheel. Miller and Valasek remotely disabled the Jeep's brakes, causing Greenberg to frantically pump the brake pedal as the vehicle drove uncontrollably into a ditch.

Vehicular surveillance

In 2023, a Reuters article exposed that Tesla was able to access camera feeds from owner's vehicles[22]. Tesla vehicles have between 8-9 cameras used for parking assistance, driver aids, and the built-in dashcam system. Touted as secure (likely secure storage, not secure access), the feeds were completely accessible to internal Tesla staff. The vehicles recorded users in compromising situations, ranging from road rage incidents to a video of a man naked in his garage. These videos were shared among staff in Tesla's offices, with some even being used as memes. In response, Tesla faces a class-action lawsuit in the United States, and China has restricted Tesla vehicles from government-owned sites. This represents a fundamental design empathy failure: users purchased vehicles believing their privacy was protected, only to discover Tesla retained unrestricted access without user consent or accessible opt-out mechanism. The cost of this design decision (loss of privacy, geopolitical restrictions) was entirely externalized to owners who had no way to anticipate or remedy it.



Original image © Tesla

▲ **Caught in 4K**

Teslas rely on an array of cameras covering the full perimeter of the vehicle for features such as Level 2 autonomous driving and Sentry Mode, a built-in anti-theft system and event recorder.

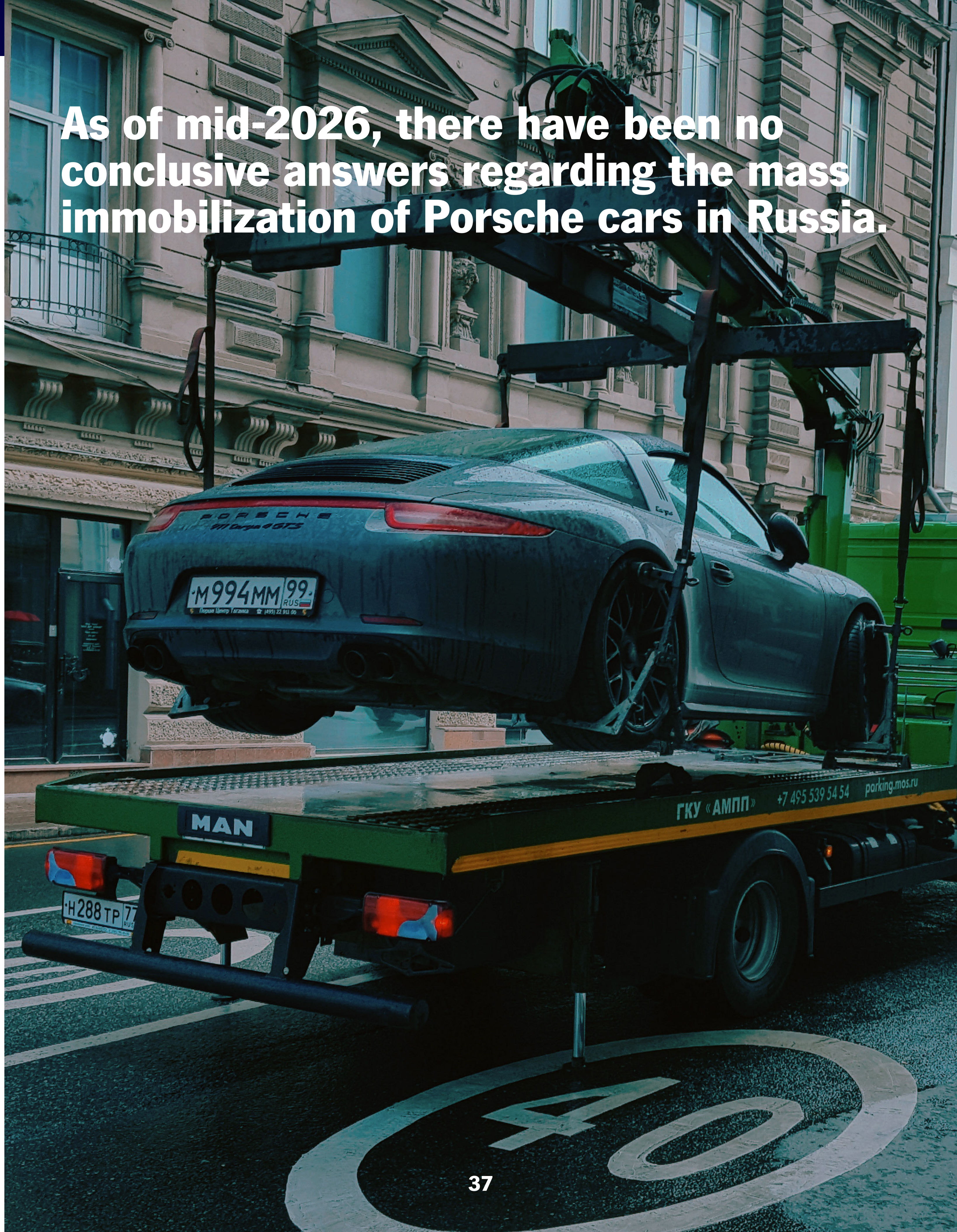
Collateral damage

In late November 2025, Porsche owners across Russia found that their vehicles would die on the road, immediately after starting, or simply not start at all, while others found that they were physically locked out of their cars entirely. There was speculation surrounding Ukrainian cyberattacks and Western kill-switches in the conversation, but per owners' forums and Russian dealership chain Rolf, the issue was with Porsche's Vehicle Tracking System (VTS), a factory-installed module on all models since 2013 utilizing satellites to protect against theft.

The Porsche vehicles affected 'panicked' when the VTS modules lost their satellite connectivity, causing them to collectively interpret the loss of connection as a theft attempt and immobilize themselves in the process. There is no official fix from Porsche as of mid-2026, while their Russian subsidiary Porsche Rusland LLC (still owned by Porsche AG due to a reported lack of buyers) has stated that an investigation is underway and a cyberattack has not been ruled out, but there are no conclusive answers yet.

Owners had to resort to guerrilla tactics to get their cars to work, trying workarounds such as disconnecting battery terminals for hours, flashing gray market hacks to the engine control unit to bypass the VTS module, to taking apart dashboards to physically extract the VTS module, effectively lobotomizing the vehicle's anti-theft system[12, 13].

> **"Currently, there is no connection for all models and types of internal combustion engines (ICEs – RBC). Any vehicle can be blocked.** Currently, the blocking can be bypassed by resetting the factory alarm unit and disassembling it. We are continuing to investigate the issue and the mechanic's options for unlocking the vehicles,"
>
> **–Yulia Trushkova,** Service Director at Rolf (translated from Russian)



Accidentally doxxed

Reliance on external cloud services to store vehicle data poses security risks previously unforeseen in the automotive sector due to the scale and nature of the data harvested by modern cars. In late 2024, an ethical hacking group accessed Volkswagen's servers and exposed the detailed ownership and location data of 800,000 vehicles across Europe.

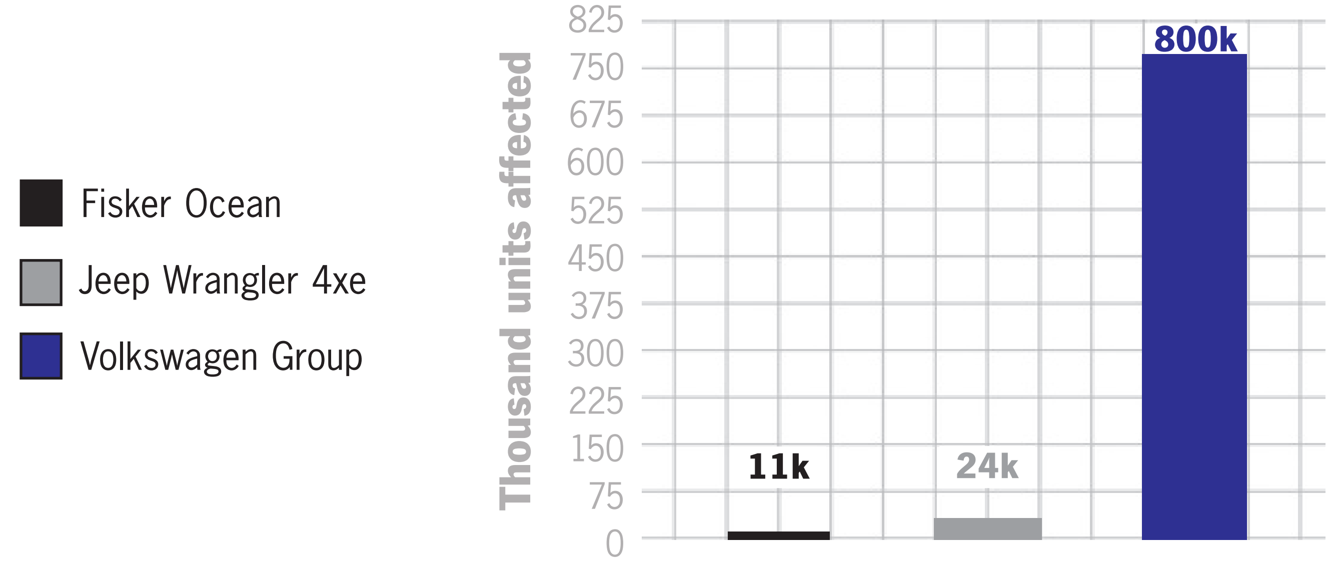
This data was insecurely stored on a misconfigured Amazon cloud managed by Volkswagen's software subsidiary CARIAD. The breach exposed GPS coordinates logged every time vehicles were turned off, in some cases revealing the precise home addresses of high-profile individuals, namely German politicians Markus Grübel and Nadja Weippert. In Grübel's case, it is reported that his car's location was logged at sensitive personal locations, military bases, and a vacation in the German region of Allgäu.

For months, anyone with knowledge of the misconfiguration could have accurately tracked specific movements of individual people[4]. CARIAD claims to have closed off access to the database soon after they were informed of the misconfiguration, but the outcome of this specific breach could have been far worse had it been conducted maliciously.

▼ Shadow diary

Vehicles recorded the location, time, and duration of each parking event. Home, work, and leisure stops are all represented, providing an accurate lens into people's lives.

> **"Much of the vehicle data can be linked to the names and contact information of drivers, owners or fleet managers.** Precise location data was viewable for 460,000 vehicles, providing hints about the lives of their owners – such as the two politicians."
>
> **–Der Spiegel International,** January 2025



▲ Public private broadcast

Der Spiegel's investigation revealed the large number of affected vehicles using CARIAD software. This breach was among the most spatially precise mass location leaks ever recorded owing to the vehicles' use of 10cm-level GPS tracking tied to people's identities.

Lucrative data mines

In a 2023 privacy research article published by The Mozilla Foundation, they concluded that cars were the worst product category they had reviewed. The variety and scale of data gathered by carmakers is wide ranging and relies on many different types of input. Buried in the fine print of their privacy policies, Kia and Nissan claimed to collect sensitive data about their buyers, including sexual and genetic data, for sale to third parties[5].

Mozilla's research found that all 25 car brands reviewed failed their basic privacy standards, a 100% failure rate worse than any other product category they have reviewed, including smart home devices, fitness trackers, and smartphones. The difference is that most consumer electronics give users some control over data collection, whereas car manufacturers provide virtually none. Manufacturers correctly anticipated that vehicles could generate valuable data and designed data collection systems that maximize extraction while minimizing user control.


Microphone


Accelerometer and gyroscope


Speedometer and tachometer


Interior attention monitoring camera


Telematics Control Unit (TCU)[G]


Infotainment system


Steering wheel

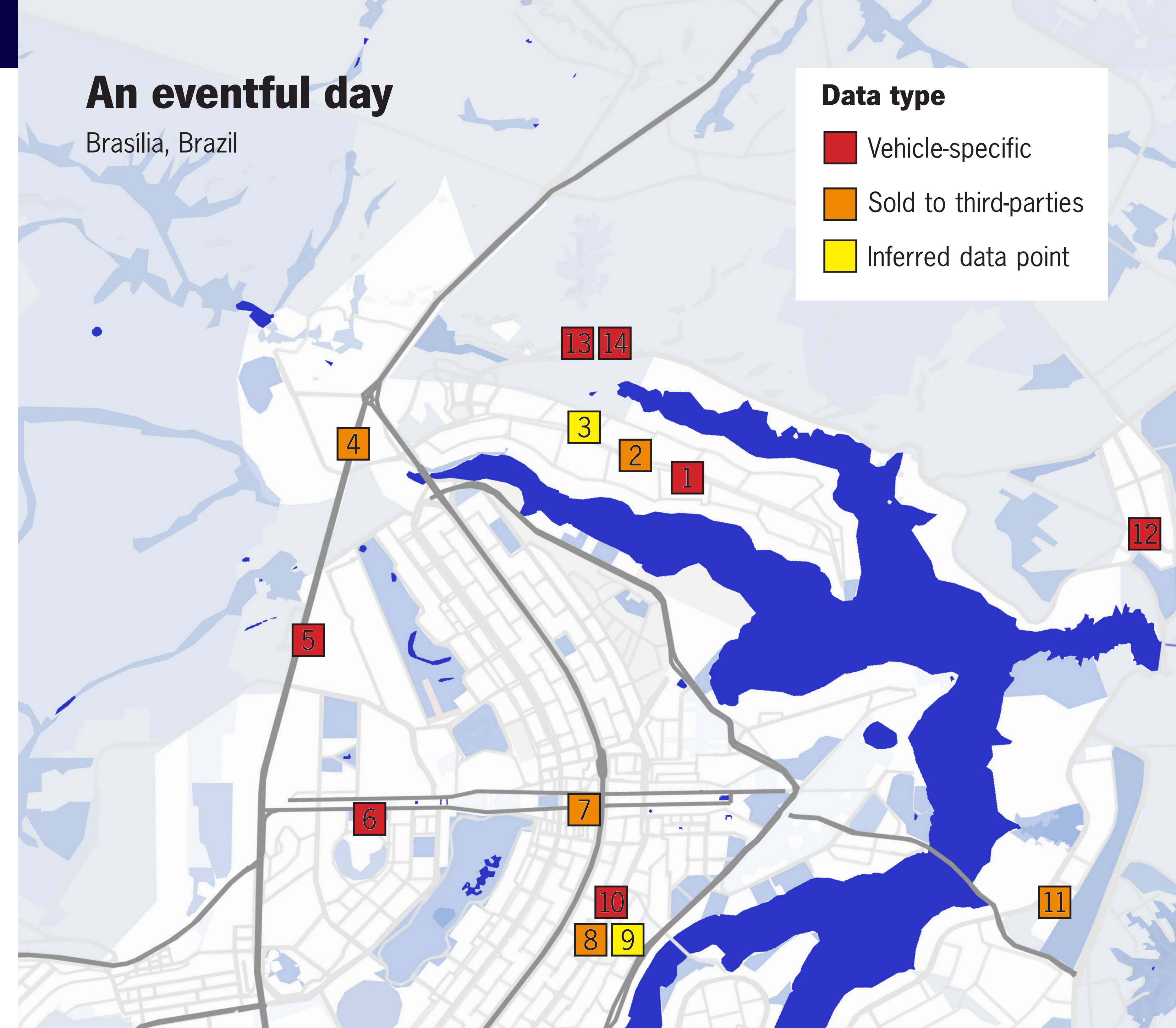

Seats

▲ **Full-stack monitoring**

Modern vehicles utilize a suite of devices that log events that occur behind the wheel and when parked. These data points are then aggregated and sold to interested third parties.

An eventful day

Brasília, Brazil



1. Car started at 21:45
2. Child detected in rear seat
3. Potential parent, detection event 6x/week
4. Voice search: "Cheap hotel"
5. Camera flags driver drowsiness
6. 21 km/h over limit, Eixo Monumental SB
7. Parked in high-crime neighborhood
8. Visited this unsaved address 4x/week
9. Left parked overnight away from home
10. Car started again at 02:05
11. 7 messages received while moving
12. Idled for 53 minutes outside hospital
13. 84% braking force, > 30° steering input
14. Airbags deployed, > 0.7g pitch reading

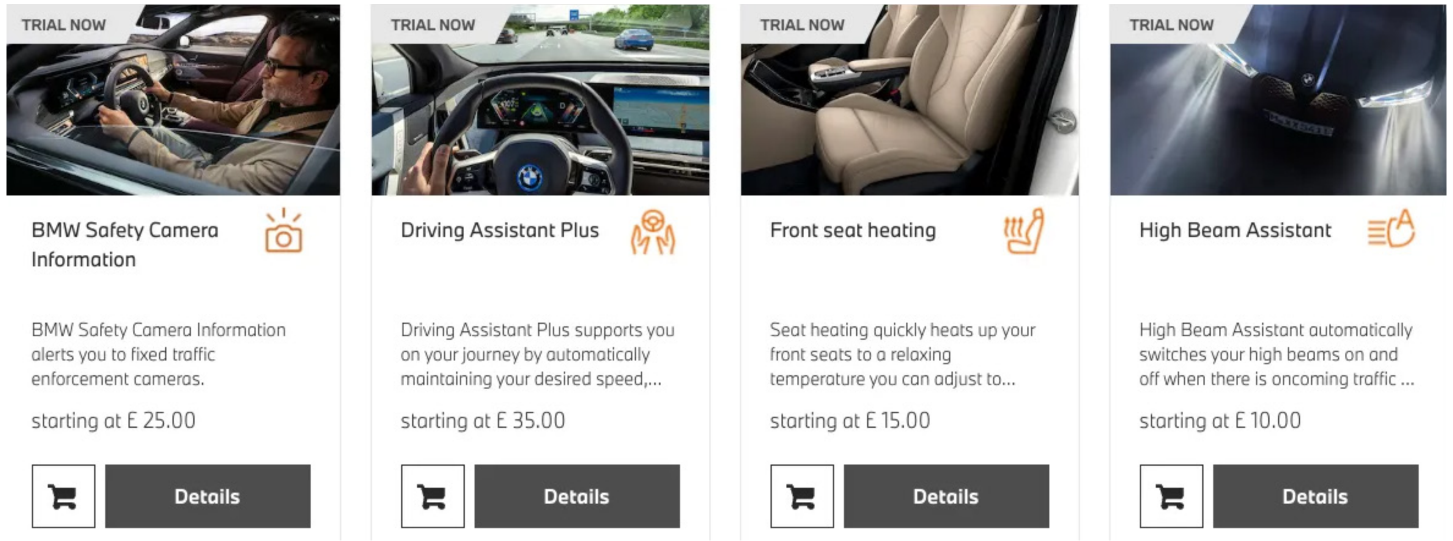
You don't own what you own

In July 2022, BMW started charging approximately $18/month for users in select markets to use their heated seats. The heated seat hardware is already installed in the car at the factory and paid for by the buyer, but BMW software-locked it until users paid a recurring fee.

Manufacturers justify subscriptions by citing ongoing server costs and OTA update development, but this argument fails scrutiny. Infrastructure costs are real but trivial compared to the revenue being extracted from paying users. Moreover, alternative models exist for features such as autonomous driving that require infrastructure and updates such as one-time feature purchases (like Tesla's "Full Self-Driving") or optional extended support plans. BMW is not alone in this practice, as Mercedes now charges subscriptions for rear-wheel steering on select models, Kia for external lighting patterns[20], and multiple manufacturers for advanced driver assistance features, while Tesla charges one-time fees for acceleration boosts[21].

When a manufacturer installs equipment or software in every vehicle but charges separately to use it, they're making two contradictory claims: that the customer paid enough to cover manufacturing and installation costs, and that the customer must pay again to use what they already bought. Both cannot be true, either the vehicle price includes the feature, making subscriptions exploitative, or it doesn't, making the installation deceptive. Severe customer backlash forced BMW to abandon hardware-based subscriptions, instead relying on them for software features such as driver assistance technology.

BMW joined a broader industry shift towards 'features-as-a-service.'
By installing hardware in all vehicles but software-locking it, carmakers can simplify manufacturing by offering fewer configurations, capture recurring revenue through subscription fees, and lock customers into lucrative ecosystems. This practice has proliferated as more manufacturers implement their own in-car subscription models.



BMW starts selling heated seat subscriptions for $18 a month [Screenshot], by M. Humphries, 2022, The Verge

▲ Hardware taken hostage

A screenshot of BMW UK's car configurator in 2022 showing options reliant on already-installed hardware such as heated seats and high beam assistance offered as subscriptions.

Original image © Mercedes-Benz
S-Class combustion variant shown, image is representative

▲ Steer-by-wire transfer

The Mercedes-Benz EQS, their top-of-the-line EV, features a complex rear-wheel steering system that reduces the car's turning radius for better maneuverability. The full functionality of this physical chassis sub-system is locked behind an annual subscription in select markets[19].

Network negligence

In February 2022, Acura's connected car service AcuraLink was discontinued for cars produced between 2014 and 2017 after AT&T sunset their 3G networks across the United States, as these cars relied exclusively on 3G modules for connectivity. As recently as July 21, 2025, AcuraLink access for models produced from 2018 to 2022 was severed even though the vehicles utilized still operational 4G LTE networks[11]. Acura did not specify a technical reason for the discontinuation, but it nonetheless resulted in the cessation of service for hundreds of thousands of vehicles produced from 2018 until 2022.

For reference, 4G LTE became commercially available in 2010 in the US, and 5G in 2019. The sunsetting of 3G was not Acura's fault, but the negligence was not planning for the demise of 3G when the network was already considered dated by the middle of the last decade. Not using other protocols as contingencies for continued communication with vehicles after the demise of a dated network is a case of structural design empathy neglect. This cessation takes an intentionally hostile turn when factoring in the July 2025 service discontinuation for vehicles produced in the past decade that utilized up-to-date telecom hardware.

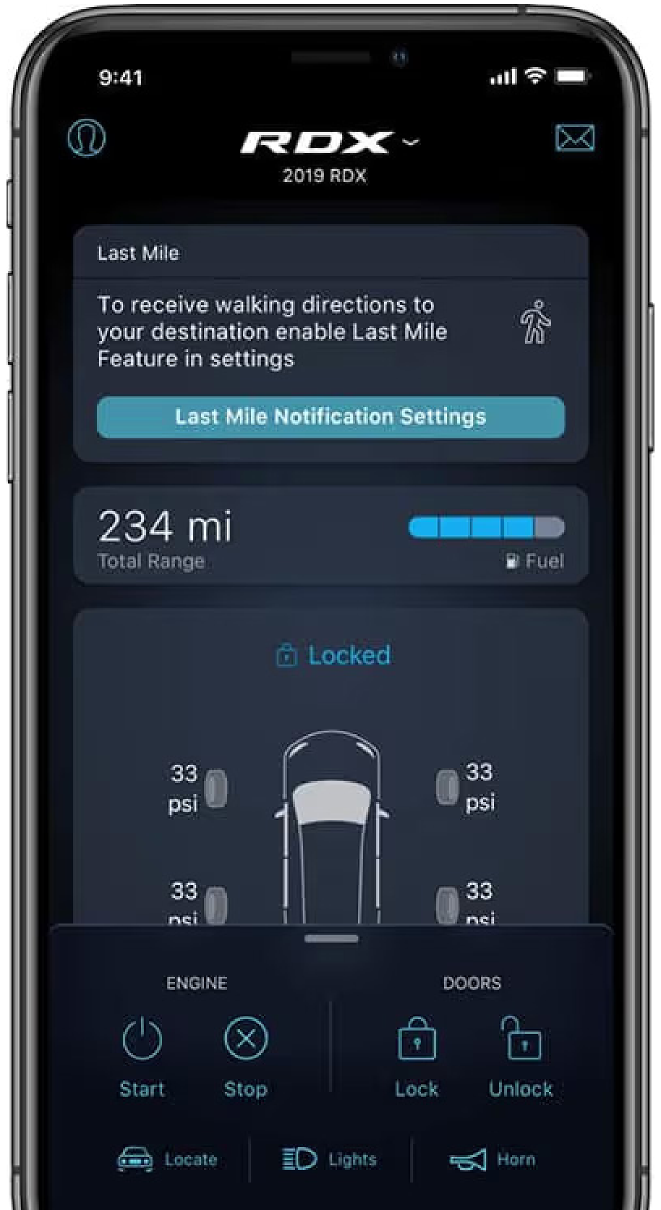
> **"What services are being deactivated for the impacted vehicles?**
> - Basic: Schedule Service, Roadside Assistance, Recall Notifications, Owner's Guide
> - Standard: Maintenance Reminders
> - Connect: Automatic Collision Notification, Emergency Call, Stolen Vehicle Locator, Enhanced Roadside Assistance, Remote Door Lock/Unlock, Send Destination, Find My Car, Destination by Voice, Vehicle Status, and Security Alarm Alert
> - Premium: Personal concierge services to make… reservations"
>
> **–Acura,** Acura MyGarage[11] (May 29, 2025), edited for brevity

Acura ILX, $30,000

Acura NSX, $170,000

▲ **Class consciousness**

Acura's cessation of 4G service across their full 2018-2022 vehicle lineup placed the $30,000 ILX sedan and the $170,000 NSX supercar in the same dispossessed league.



Original image © Acura

◀ **You're on your own**

The discontinuation of AcuraLink has eliminated connected features on cars under a decade old with little recourse for owners. Acura has not provided any reasoning behind the decision to cut off products such as the RDX that were still actively in production.

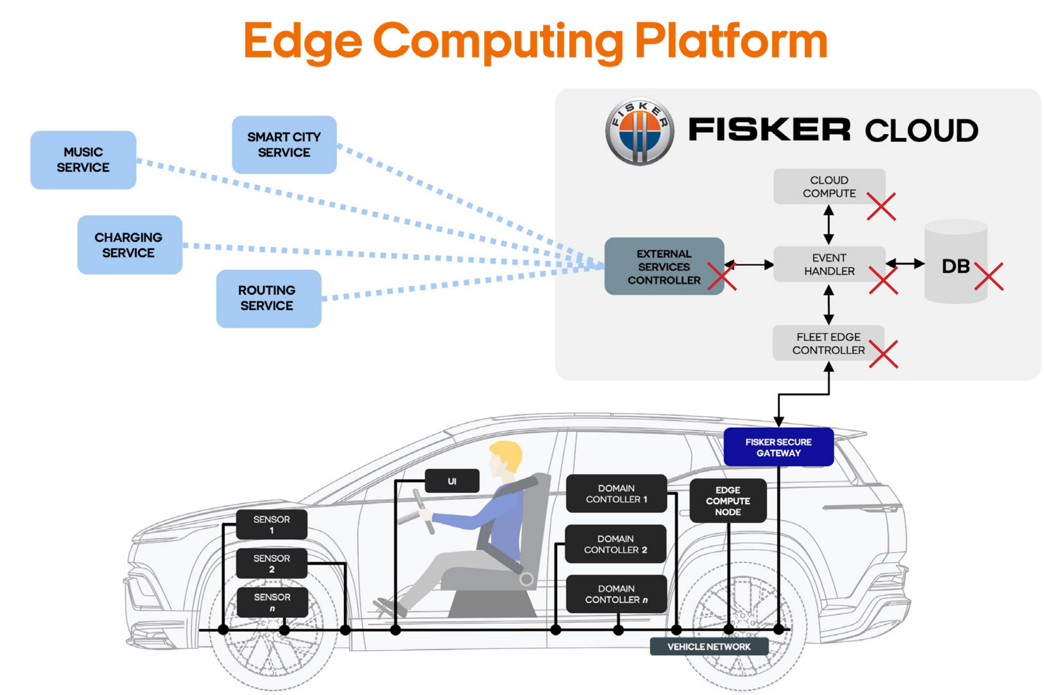
External Dependencies

The Fisker Ocean, a cutting-edge electric SUV first unveiled in early 2020, signaled Fisker's return to market, but failed spectacularly soon after launch. Poor management drove Fisker to file for Chapter 11 bankruptcy in June 2024, degrading customers' cars in the process. The cars had software-locked features and used proprietary diagnostics software that only their technicians could access, so without Fisker as a company, these support networks disappeared. There was no degradation plan or a way for owners to maintain software-dependent features independently, meaning vehicles already plagued with software bugs lost core SDV functionality such as navigation, OTA updates, and remote app features.

> **"Through edge computing and 4G, later ultra-low latency 4G connectivity, it also becomes possible for cloud computing resources to be used as a seamless extension of the computing power in the car.** Continuous software updates, both for embedded systems in the car and functions hosted in the cloud, let the digital car grow and become smarter over its lifetime... We are designing our EVs to always be 'connected'. Our next-generation connectivity platform is already heavy at work seamlessly integrating online services and functions, Fisker-unique as well as third party services... With data analysis, cloud computing, and the ability to push OTA updates to the vehicle, we expect the in-car experience will evolve over time..."
>
> **–Fisker Inc.** SEC filing Accession No. 0001720990-24-000045, edited for brevity

Fisker produced approximately 11,000 Ocean SUVs between 2023-2024. When a company purchased Fisker's assets in bankruptcy, they initially agreed to provide ongoing support, then reneged on that commitment, resulting in the Fisker Owners Association filing a lawsuit in November 2024 seeking access to proprietary software and infrastructure.

This rendered the entire production run of 11,000 mechanically sound cars artificially obsolete due to software lockout. Fisker's design decisions assumed the company's perpetual existence, in contrast to Saab, a Swedish automaker that filed for bankruptcy in 2011 after 62 years of manufacturing. Despite the company's collapse, Saabs remain viable vehicles because they operate independently of the company that made them. There is no external dependency, and enthusiast communities share tribal knowledge that keeps cars on the road. When Fisker failed, there was no part to remove or forum knowledge to apply that could restore lost functionality. In late 2025, third party company OV Loop stepped in to provide software support via a new $499 annual subscription, transferring the dependency rather than resolving it.



Fisker Inc. (2024). Form 10-K. U.S. Securities and Exchange Commission.

▲ **Falling from cloud nine**

Fisker's SEC filing from fiscal year 2023 describes their cloud infrastructure used for vehicle functions, with multiple dependencies requiring Fisker's continued operation[1]. When Fisker filed for bankruptcy in 2024, this entire ecosystem collapsed.

Fisker's SEC filings exhibit awareness of financial precarity and the risks their connected architecture posed to owners.

"Interruption or failure of our information technology and communications systems could impact our ability to effectively provide our services. We outfit our vehicles with in-vehicle services and functionality that utilize data connectivity to monitor performance and timely capture opportunities for cost-saving preventative maintenance. The availability and effectiveness of our services depend on the continued operation of information technology and communications systems, which we have yet to fully develop."

–Fisker Inc. SEC filing Accession No. 0001720990-24-000045, edited for brevity

"Our systems will be vulnerable to damage or interruption from, among others, fire, terrorist attacks, natural disasters, power loss, telecommunications failures, computer viruses, computer denial of service attacks, cyber attacks or other attempts to harm our systems. In addition, our vehicles are highly technical and complex and may contain errors or vulnerabilities, which could result in interruptions in our business or the failure of our systems."

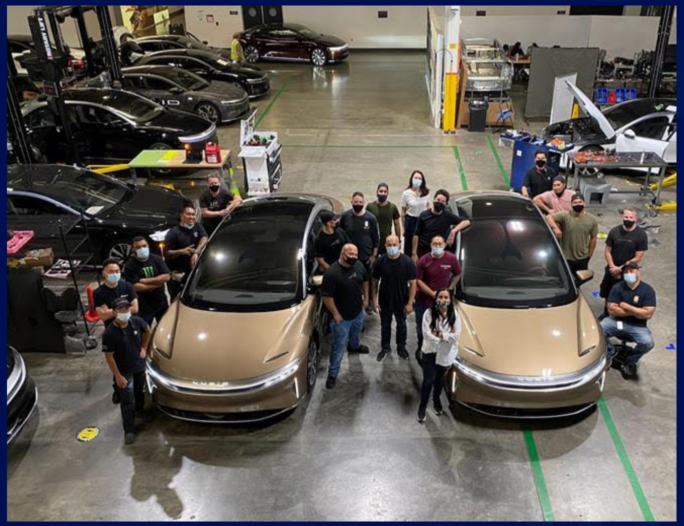
–Fisker Inc. SEC filing Accession No. 0001720990-24-000045, edited for brevity

"Operational Risks:
There is substantial doubt about our ability to continue as a going concern."

–Fisker Inc. SEC filing Accession No. 0001720990-24-000045

Industry expert Q&A

▲ **Lynette Cox**
EV Powertrain Systems Engineer



▲ **Winners' circle**
Pictured: Lynette with her development team standing alongside their award-winning vehicles.

To understand automotive design decisions from an expert perspective, this publication includes insights from Lynette Cox, a senior EV Powertrain Systems Engineer who has worked on vehicle development at two prominent American EV manufacturers since 2021. Her experience spans from early-stage vehicle development through production, providing nuanced context for the pressures shaping modern automotive design and engineering.

Interview responses represent Lynette's own personal opinions and are not a reflection of her employers' views or processes.

Responses have been edited for clarity.

How much of a modern vehicle's core functionality requires cloud connectivity?

In my experience, none of a vehicle's core functionality requires cloud connectivity. A vehicle cannot and should not be left stranded because of a lost internet connection. One could argue that OTA software updates (which can include modifications to core vehicle behavior) have a cloud dependency. Systems like body controls managing your frunk and doors, battery management systems for both high and low voltage components, the charging system responsible for communicating with EVSEs (Electric Vehicle Supply Equipment) and enabling energy flow into the batteries, the powertrain systems delivering torque to the wheels, thermal management for the battery, motor, and cabin do not depend on an internet connection. They run on embedded processors integrated directly into the vehicle, making the cloud a convenience layer.

What's an example of a design decision that looks good on paper but creates real-world frustration for EV owners?

OTA updates are a great example of a feature that sounds transformative in theory. If a manufacturer wants to introduce a new feature or fix a software issue, it can be pushed to the vehicle with a simple button press with no dealership visit required. In practice, the convenience of connectivity is entirely dependent on having a reliable internet connection. If your EV is parked somewhere with poor connectivity like underground parking, you may not just lose the ability to receive an update, but also lose access to the mobile app entirely, meaning you can't precondition the cabin before a long drive or even check your state of charge. The same features designed to make ownership seamless become unavailable when a driver might need them most.

What features do you scrutinize most when evaluating an EV?

Having had the privilege of driving a number of electric vehicles, there are a few areas where I always pay close attention, because how a vehicle performs in these areas speaks to the quality of the engineering behind it.

First is thermal management. A vehicle's cabin comfort and cooling systems need to be robust across the full spectrum of real-world conditions from the coldest Alaskan winters to the peak heat of a Texas summer.

What makes this particularly challenging is that battery cells are highly sensitive to temperature (too cold and they lose efficiency and charge acceptance, too hot and you're actively degrading cell health and longevity). The thermal system has to balance keeping the cells in their optimal operating window while simultaneously managing cabin comfort, and it has to do that under wildly varying environmental conditions. If range drops significantly because you happen to live somewhere with extreme temperatures, or if the cooling system struggles to keep up on a hot day, that's a meaningful engineering shortcoming. In the most extreme cases, like charging outside during a blizzard, I've seen situations where the energy required just to maintain safe battery temperatures can outpace that of the charger.

This points to what I'd argue is the single biggest vulnerability of EV ownership: the high voltage battery. If a cell's voltage drops to or below its minimum threshold from being stored without charging for an extended period or from an extreme thermal event, the battery cannot recover. You're not looking at a software fix or a minor repair; but instead a full battery replacement. For a component that represents a significant portion of the vehicle's value, that's a serious consequence of what

This points to what I'd argue is the single biggest vulnerability of EV ownership: the high voltage battery. If a cell's voltage drops to or below its minimum threshold from being stored without charging for an extended period or an extreme thermal event, the battery cannot recover.

Second is the charging experience. Charging is one of the most frequent interactions a driver has with their vehicle, so it needs to be seamless. An error telling you to unplug and reconnect is a small moment of friction that reflects a much deeper engineering challenge. In my experience working closely with charging systems, every EVSE manufacturer interprets and implements written charging standards slightly differently, and hardware designs can vary, too. That means a single vehicle charging system has to be robust enough to handle all of those real-world variations flawlessly, while still delivering the fastest and most efficient charge possible. That requires enormous amounts of testing across a wide range of charging hardware in the field. When the charging experience is smooth, it's easy to take for granted, but when it isn't, it's apparent. To me, it's one of the clearest indicators of how seriously a team invested in system validation.

Finally, low-speed torque management and braking. This one I've been thinking about a lot lately after spending time in a newer EV. Electric motors are inherently harder to control at low speeds, and there's also the matter of backlash in the gear train – essentially a slight spring-like compliance that becomes noticeable at the moment the vehicle comes to a complete stop. Without the right controls and software calibration to compensate, the vehicle will lurch subtly as the motor reaches zero speed. It's a small thing, but it's noticeable on every single stop, and it's the kind of detail that well-executed software can eliminate entirely. When I feel it, I know it's a piece of refinement that didn't make it across the finish line.

What's a common tension between engineering and product teams during vehicle development?

It really depends on how a company is structured and how much engineers' voices are heard in the design process. In my experience, the two teams genuinely need to meet in the middle. Engineers bring critical technical constraints to the table (legally required notifications, for example) while designers understand how to translate those requirements into an intuitive UX.

My view is that user experience should be designed first around the vehicle's core feature set, informed by consumer data and feedback, and then engineering input should be layered in to pressure-test and refine until features reflect the best possible experience from concept through delivery!

I've been in plenty of spirited discussions about something as specific as warning message language. Engineers naturally gravitate toward precision: if something goes wrong, they want the notification to say exactly what failed. Designers, rightly, push back in favor of clarity for the average driver. Neither instinct is wrong, as the tension is productive when both sides are heard.

My view is that user experience should be designed first around the vehicle's core feature set, informed by consumer data and feedback, and then engineering input should be layered in to pressure-test and refine until features reflect the best possible experience from concept through delivery!

Principles for tomorrow

The failures documented in this paper are the compounded result of design decisions that prioritized short-term business outcomes over the people who live with those decisions for years. The engineering talent to create durable goods exists, but the incentive to do so without external pressure is questionable. The following principles are actionable architectural and design decisions that require marginal additional investment relative to the harm their absence causes.

Design for independence

Think of dependencies as a leash tying an asset to its maker; if the maker goes under, the product follows suit. Dependencies don't make a product smart, they make them vulnerable. Product independence goes farther than simply designing something to work offline, it requires an architecture of ownership that facilitates independence at the foundational level.

For example, complementary applications that allow remote interaction with modern vehicles rely on server access and ongoing maintenance like any other application does; they are fundamentally no different because they need continuous maintenance from developers to ensure continued support and operation after launch. Graceful degradation[G], defined as designing systems to fail safely and partially rather than completely and catastrophically, is standard practice in aerospace and medical device engineering and should extend to the vehicles that people depend on.

Recent events show that the long term viability of software features with external dependencies is contingent on the quality of the framework supporting them. This is where local protocols make sense as a graceful degradation fallback, or even as a dedicated privacy-focused architecture. While range-limited when compared to cellular or internet connectivity, a local/peer-to-peer (P2P)[G] protocol such as Bluetooth LE (BLE)[G] or LoRaWAN (Long Range Wide Area Network)[G] can still be secure, effective, affordable, and independent.

Mesh networks[G] are widely used within and beyond the auto industry. For instance, Volvo uses private LoRa tracking tags to locate built trucks within their plants and shipping yards. Philips uses a mesh network to facilitate communication between Hue lightbulbs. Finally, Amazon's Sidewalk mesh network allows Echo devices and Ring cameras to form a neighborhood-wide network that keeps them connected even when a homeowner's primary internet goes offline

The Apple AirTag proves the viability of offline connectivity by maintaining connected features without an omnipresent internet or data connection. By anonymously relaying their locations using BLE through a decentralized mesh of nearby iPhones (that then relay AirTag locations to iCloud), the need for direct internet connectivity to the AirTag is sidestepped. Apple's integration requires their own network of devices, but open source options exist.

Unlike cellular connectivity which requires a contract with a telecom provider, local protocols like BLE and LoRaWAN are private and permissionless. While these protocols evolve over time, the backwards compatibility standard of Bluetooth has been proven over decades, whereas cellular networks have a hard expiration date. Manufacturers should utilize similar decentralized protocols instead of storing attributable data on vulnerable servers.

Design for the long tail

A vehicle can only be as reliable as the first difficult to service part that breaks. In order to live up to the historic expectation of multi-generational ownership, failures should be expected, and remedies designed with intention. The practice of requiring major disassembly for minor service procedures is unrealistic and effectively functions as a financial and logistical gatekeeper to longevity, paring down the pool of surviving products to only those whose owners could afford repairs. Forcing products out of use through engineered attrition is bad for a brand's legacy and harmful to our environment.

The architecture that underpins cloud-based connectivity exists beyond any individual's control. If a company hosting cloud-based software goes bankrupt like Fisker did, any artificial barricades on already-purchased items function as de-facto blockers from normal operation. These barricades assume perpetual existence that should never be considered as a given, which is why user agency should be designed-in at the foundational level with right-to-repair, in turn allowing products to last longer.

The VR6's longitudinal application with plastic cooling pipes, Fisker's brittle connected systems architecture, and AcuraLink's archaic 3G dependency all share a lack of foresight. Small off-the-shelf alterations to existing designs (i.e. metal pipes instead of plastic for the VR6, open-source diagnostics for the Fisker Ocean, and a more advanced communication protocol such as the addition of a 4G module for older AcuraLink-equipped vehicles in the last decade) would have prevented otherwise functional systems from an untimely demise.

Just as a computer allows for the upgrading of network cards, RAM, and processors without replacing the entire motherboard, a vehicle designed and built with a 20 year service life cannot exclusively rely on technological standards that become obsolete within a decade; the individual modules that enable a vehicle's software features should be decoupled from the vehicle's core operating system, allowing for modularity and eventual upgradability that stands the test of time.

Anticipating the long tail entails an awareness of the weakest link of any system, whether mechanical stress or technological obsolescence, and designing robustness into it based on the context in which it operates. The architecture of a well-designed SDV already accounts for this given that decoupling hardware from software is the stated goal of modern vehicle development, allowing individual modules to be updated or replaced without compromising core functions. The failures documented in this paper are not a consequence of the SDV itself, but of implementations that coupled critical functions to external infrastructure that was not guaranteed to operate reliably.

Beacon concept study

Beacon is a permissionless keyfob that eliminates the third-party server dependency built into modern connected vehicles. Developed conceptually in conjunction with this publication, Beacon is designed as a response to the growing inability of people being able to take ownership of their devices.

Instead of routing commands via a mobile app through a manufacturer's cloud infrastructure, Beacon communicates directly with the vehicle over local and secure Ultra-wideband (UWB)[G] and LoRaWAN protocols operating on Class C (always-on) modules to remotely complete functions such as preconditioning the cabin and monitoring vehicle vitals.

Communication: UWB + Class C LoRaWAN module
Range: Primarily-local, long-range (LR) capable
Power supply: Thermally stable sodium ion (Na-ion) battery, user serviceable
Dependencies: No manufacturer dependencies, LR relies on existing LoRaWAN network
Form factor: Puck, 40mm diameter, recycled plastic chassis
Cybersecurity: UWB utilizes Time-of-Flight (ToF)[G] tracking for added security
Charging: In-vehicle wireless charging and USB-C port

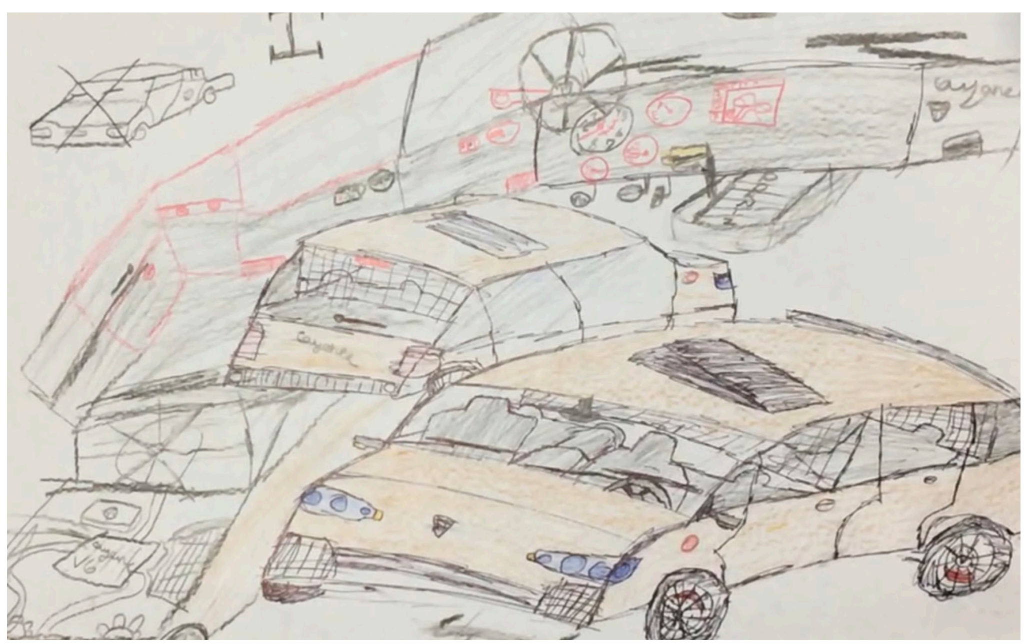
This product concept is designed to be pocketable, repairable, resilient, and completely independent; a result of architecting core functions to operate out of walled environments.

Author's note

In addition to this paper's industrial critique, it also functions as a defense of curiosity and awe held by a younger version of myself. Growing up in the rapidly changing 2000s, this machine of seemingly infinite capability left a lasting impression that shaped my design ideals.

I wonder about not the future of that outlook, but the ability to act on it. Lately, it seems that outlook is dimming as companies bolster their bottom line at a human cost that will eventually have consequences. While this disregard is not a new phenomenon, the rapid and decentralized proliferation of it is. We are currently at a crossroads that will define the next generation of ownership beyond the automobile, and whether these industrial experiments become pervasive norms at our expense. The things we build are a reflection of the values we hold; valuing the craft of building products to a high standard builds trust, and that trust is worth defending.

▶ **Porsche Cayenne**
by Bearing Studio, 2008



About the author

◀ **Zane Zahran**
Industrial designer, Red Team Analyst

Zane Zahran is a multidisciplinary designer based in Austin, Texas. His design philosophy is a product of his experience red teaming artificial intelligence models for a living, as well as growing up in Kuwait, Egypt, and Texas; places that value robust engineering and autonomy.

His focus on hostile design crystallized in 2022 after a failure of his 2017 MacBook Pro's CPU, which is soldered to the logic board. This led him to the Austin-based shop of seasoned right-to-repair advocate Louis Rossmann, who diagnosed the failure and explained the $700 bill that comes with an otherwise unnecessary logic board replacement.

A few years later, his experience repairing a broken Porsche Cayenne showed him that this disregard for users has existed in less evolved forms. Through this background, he presents this publication as an audit of how disposable logic is re-engineering the automobile.

Glossary

Technical Terms

BLE (Bluetooth Low Energy)
A short-range wireless communication protocol designed for low power consumption. Unlike cellular connectivity, BLE operates without a telecom contract or internet dependency, making it a viable privacy-respecting alternative for local vehicle-to-device communication that does not rely on manufacturer infrastructure.

CANBUS (Controller Area Network Bus)
A communication system that allows the various electronic control units within a vehicle to exchange data without a central computer. It functions as the vehicle's internal nervous system; damage or water intrusion near CANBUS components can produce cascading, difficult-to-diagnose failures across unrelated systems.

Longitudinal vs. Transverse Engine Layout
Longitudinal engines are mounted front-to-back (typical in rear-wheel-drive vehicles), whereas transverse engines are mounted sideways (typical in front-wheel-drive vehicles). The Cayenne's VR6 was adapted from transverse to longitudinal, creating serviceability hurdles.

LoRaWAN (Long Range Wide Area Network)
A low-power wireless protocol capable of transmitting small amounts of data over long distances without cellular infrastructure. Its permissionless, decentralized nature makes it a candidate for vehicle connectivity applications that prioritize owner independence over manufacturer data collection.

Mesh Network
A network where devices relay data for each other rather than routing through central infrastructure, so the network survives the loss of any individual node.

NVH (Noise, Vibration, Harshness)
Engineering term describing the undesirable sensory feedback from vehicles during operation. Reducing NVH is a key design goal, often requiring trade-offs with other priorities like cost or serviceability.

OTA (Over-The-Air)
Software updates delivered remotely to products via internet connectivity, allowing manufacturers to fix bugs, add features, or modify product behavior without requiring physical service visits.

P2P (Peer-to-peer)
A communications architecture where devices interact directly with each other rather than routing data through a central server. In the context of vehicle connectivity, P2P protocols offer an alternative to cloud-dependent systems that become non-functional when manufacturer infrastructure is discontinued or compromised.

SDV (Software-Defined Vehicle)
A vehicle where core functions such as propulsion, braking, and interface are controlled primarily through software rather than mechanical systems.

TCU (Telematics Control Unit)
The hardware module responsible for a vehicle's wireless communications, enabling features such as remote diagnostics, OTA updates, emergency calling, and data transmission. The TCU is the primary vessel through which manufacturers collect vehicle and user data.

ToF (Time-of-Flight) and UWB (Ultra Wideband)
Short-range radio protocol that locates devices by precisely timing signal travel (Time-of-Flight), making relay and spoofing attacks far harder than with conventional key fobs

Design & Industry Concepts

Cloud dependency
Architectural design where product functions require continuous connection to manufacturer servers. Creates single point of failure if servers are decommissioned.

Graceful degradation
The design principle that a system should fail partially and safely rather than completely and catastrophically. A product engineered for graceful degradation continues to provide core functionality when non-critical components fail, rather than becoming entirely inoperable. Standard practice in aerospace and medical device engineering; largely absent from current automotive software design.

Planned obsolescence
The deliberate design of a product with a limited lifespan, ensuring it becomes outdated or non-functional within a predictable timeframe. This can be achieved through material choices, software dependencies, discontinued support, or the withholding of repair resources.

Right-to-repair
Movement and legislation requiring manufacturers to provide consumers and independent repair shops with access to tools, parts, and information needed to service products independently.

Subscription features
Business model where hardware is installed in vehicles but software-locked until users pay recurring fees. Example: BMW heated seats requiring monthly payment.

Telemetry
The automated collection and transmission of data from a remote source to a receiving system for monitoring and analysis. In modern vehicles, telemetry refers to the continuous transmission of operational, behavioral, and location data from the car to manufacturer servers — often without explicit user awareness or meaningful consent mechanisms.

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